LESSON 7.8

Ordered Pairs in All Four Quadrants

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

WHY THIS MATTERS Captain Vega realized her treasure map only covered one corner of the island! She needs to expand the map in all directions — north, south, east, and west of the base camp at the origin (0, 0). Now clues can have negative coordinates: (-4, 3) means 4 units west and 3 units north. A clue at (2, -5) means 2 units east and 5 units south. Which quadrant is each clue in?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Quadrant *Spanish: Cuadrante*	One of the four parts of a coordinate grid, numbered I to IV.	I (+, +) top-right, II (-, +) top-left, III (-, -) bottom-left, IV (+, -) bottom-right
Negative coordinate *Spanish: Coordenada negativa*	A coordinate less than zero. It means left or below the center point.	(-3, 2) means 3 left and 2 up from the origin
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the y-axis is (-3, 2) — same distance, opposite side
Axis *Spanish: Eje*	One of the two number lines that frame a graph: the x-axis runs across, the y-axis runs up and down.	The x-axis goes left-right; the y-axis goes up-down; they cross at (0, 0)
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I will plot (-3, -4). I start at the origin (0, 0).

1. The x is -3, which is negative, so I move 3 units to the LEFT.
2. The y is -4, which is negative, so I move 4 units DOWN. I place my dot.
3. Both coordinates are negative, so the point is in Quadrant III.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot (-5, 2). The x is -5, so which way do we move? Left 5 units.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In Quadrant _____, the x-coordinate is _____ and the y-coordinate is _____. I know (-3, -4) is in Quadrant _____ because both coordinates are _____.

Word bank Quadrant | Negative coordinate | Reflection | Axis | Integer | quadrant | positive | negative | x-coordinate | y-coordinate | left | right

WATCH OUT!

A common mistake

A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis

LESSON 7.8

Ordered Pairs in All Four Quadrants

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each ordered pair to its quadrant.

_____ (3, 4)	(A) Quadrant I
_____ (-2, 5)	(B) Quadrant II
_____ (-6, -1)	(C) Quadrant III
_____ (1, -3)	(D) Quadrant IV
_____ (-4, 2)	(E) Quadrant II
_____ (5, -6)	(F) Quadrant IV

HINT Only the SIGNS matter: (+, +), (-, +), (-, -), and (+, -) each belong to one quadrant.

2 **Write the name of the correct group on each line.**

Order the points by quadrant: I, II, III, IV. Within the same quadrant, put the point with the greater x-coordinate first.

Groups:

_____ (-3, -2)

_____ (5, 1)

_____ (-1, 4)

_____ (2, -6)

_____ (4, 3)

_____ (-5, -7)

HINT Use the signs: (+,+) is I, (−,+) is II, (−,−) is III, and (+,−) is IV.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

In which quadrant is the point (-4, 5)?

(A) Quadrant I
(B) Quadrant III
(C) Quadrant II
(D) Quadrant IV

HINT Look at the signs: x = -4 is negative and y = 5 is positive.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

(A) Quadrant I
(B) Quadrant II
(C) Quadrant IV
(D) Quadrant III

HINT Positive x pushes you RIGHT; negative y drops you DOWN.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which point is located on the x-axis?

(A) (-4, -4)
(B) (0, 4)
(C) (4, 0)
(D) (4, 4)

HINT A point ON the x-axis never moves up or down — its height is zero.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** Name the quadrant for the point (-2, 5).
2. **Student thinking** The x is -2, so I move 2 units DOWN. The y is 5, so I move 5 units to the RIGHT.
3. **Student answer** (-2, 5) is in Quadrant IV.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Which coordinate controls left/right, and which controls up/down? The x is always the left/right move.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

SENTENCE STARTER In Quadrant ___, the x-coordinate is ___ and the y-coordinate is ___. I know (-3, -4) is in Quadrant ___ because both coordinates are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.8

Ordered Pairs in All Four Quadrants

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

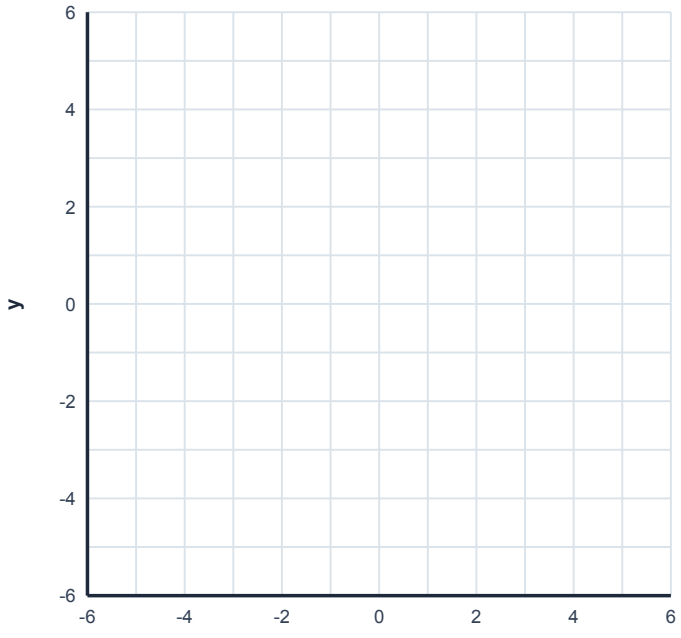
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot one point in each quadrant: (3, 2), (-4, 1), (-2, -3), (5, -4)



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

For each ordered pair, write the signs of x and y and name the quadrant (or axis).

Point	x sign	y sign	Quadrant or Axis
(6, -2)			
(-3, -5)			
(0, 7)			
(-1, 8)			
(4, 0)			

SHOW YOUR WORK

3 **Write the letter of the matching item on each line.**

Match each quadrant to its coordinate sign pattern.

_____ Quadrant I
_____ Quadrant II
_____ Quadrant III
_____ Quadrant IV
_____ x-axis
_____ y-axis

(A) (+, +)
(B) (-, +)
(C) (-, -)
(D) (+, -)
(E) y = 0
(F) x = 0

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Point A is at (-5, 3). If you move 8 units to the right, what are the new coordinates?

(A) (-13, 3)
(B) (-5, 11)
(C) (-5, -5)
(D) (3, 3)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

In which quadrant is the point (-3, 5)?

(A) Quadrant I
(B) Quadrant III
(C) Quadrant IV
(D) Quadrant II

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** Name the quadrant for the point (6, -1).
2. **Student thinking** I'll read the coordinates as (y, x), so I treat this like (-1, 6): the first number -1 is negative (left) and the second number 6 is positive (up).
3. **Student answer** (6, -1) is in Quadrant II.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.8

Ordered Pairs in All Four Quadrants

Challenge · Extension

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

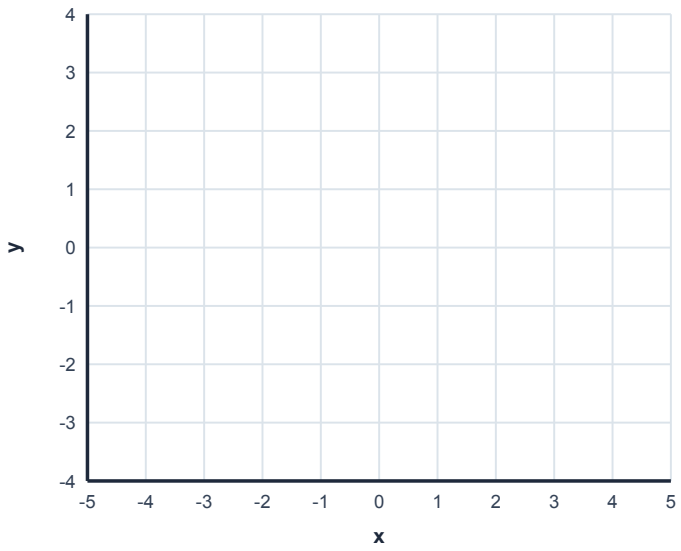
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot these 4 points and connect them in order to form a shape. Name the shape and identify which quadrants it spans: (-3, 2), (3, 2), (3, -2), (-3, -2)



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A point (a, b) has a < 0 and b < 0. In which quadrant does it lie?

(A) Quadrant I

(B) Quadrant II

(C) Quadrant IV

(D) Quadrant III

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Reflecting the point (3, -6) across the x-axis gives which point?

(A) (-3, -6)

(B) (-3, 6)

(C) (3, 6)

(D) (6, 3)

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Find the mistake. Explain it, then show the correct work.**

Find the Quadrant Error

1. **Problem** Identify the quadrant for (-3, -5)
2. **Student thinking** The x is -3, so I go left 3. The y is -5, so I go up 5.
3. **Student answer** (-3, -5) is in Quadrant II

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 **Answer in complete sentences.**

A point starts in Quadrant I at (4, 6). Describe a move that puts it in Quadrant III. What must happen to both coordinates? Can you give the exact new coordinates after your move?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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